Memorized Text Lives in the Flat Directions: A Two-Order Theory of Curvature-Based Weight Editing

Anonymous

Abstract

Curvature-based weight editing removes memorized text from a language model by deleting or shrinking weight components ranked by an approximate Hessian. In practice no approximation dominates, and a plain identity metric is hard to beat. We explain why, and we say what does move the trade-off. Seen through the loss curvature of a reference population, a trained MLP matrix has a small *head* of high-gain directions and a large, near-degenerate *bulk*. In four models from two families (OLMo-2 7B and 1B, OLMo-3 7B, Qwen2.5-7B) we find that memorized text is stored in the bulk to a degree that varies by population and predicts what an edit can do: populations memorized during pretraining are strongly *whitened* relative to ordinary text on both sides of the matrix and are forgotten when the bulk is deleted, while globally duplicated boilerplate memorized late in training is half as whitened, survives the same deletion, and is reached only by a coherent step along the bulk direction its items share. A theory in two Taylor orders of the token loss, whose only object is the population Fisher, accounts for these facts and makes parameter-free predictions that hold in two models with identical code: the collateral of block edits within 2–10%, the margin-shift variance of a layer perturbation within a factor two and its ratio between populations within 30%, and forgetting curves from per-token margins. A minimum-interference argument gives the storage cost of an item as the inverse product of its whitened energies on the two sides, which predicts that flat items are cheap to store and that learning should flatten what it memorizes by redistribution. A causal experiment tests the mechanism: a fresh imprint written by SGD follows the item gradient and is head-heavy, one written by the natural gradient is in the bulk and has the profile of the pretraining-era populations, the items' representation recodes toward the bulk as they are fit whatever the optimizer, and under continued population training the head-heavy imprint is erased first while every imprint's recoded representation drifts back, so persistence needs recurrence, not location alone. Inside the bulk the cost of a label-free edit depends only on the energy it moves, so curvature fidelity cannot matter there, which is why editing methods share one frontier; placement across layers and coherent set-level directions are the levers that move it, and the collateral of a coherent edit is the alignment of its direction with the preserved population.

1 Introduction

Large language models reproduce training text verbatim [Carlini et al., 2023, Nasr et al., 2023], and a growing family of methods removes such text by editing weights rather than by retraining [Eldan and Russinovich, 2023, Jang et al., 2023, Maini et al., 2024]. One line of work ranks weight components by an approximate loss Hessian, on the intuition that memorized content and general capability occupy different parts of the curvature spectrum, and deletes the components with low

Preprint.

curvature [Merullo et al., 2025, Anonymous, 2026]. The approximations in use, K-FAC [Martens and Grosse, 2015], its eigenvalue-corrected form [George et al., 2018], Shampoo [Gupta et al., 2018] and others, differ in fidelity, and one would expect the more faithful ones to edit better. They do not: their trade-off frontiers between forgetting and retained capability coincide, and an identity metric, which ignores curvature altogether, sits on the same frontier [Anonymous, 2026]. Nothing in the intuition explains this, nor says what the frontier is, nor whether anything can move it.

This paper gives the missing account. Its object is a single MLP weight matrix seen through the curvature of the token loss on a reference population, in the Kronecker-factored eigenbasis, whose spectrum has a small *head* of high-gain directions and a large, nearly degenerate *bulk*. We make three contributions.

Laws (Section 3). In four models from two families, memorized text is written in the bulk to a degree that is a property of the memorized population and that predicts what an edit can do. Populations memorized during pretraining are *whitened* relative to ordinary text, their activation and gradient energies per direction falling as a power of the reference energy on both sides of the matrix, and deleting the bulk block forgets them at the price of noise of the same norm. Globally duplicated boilerplate that the same models memorized in the last 50B tokens of training is half as whitened and survives the same deletion. Every memorized population shares a bulk direction of its own, orthogonal across populations and to what ordinary text uses. Removing the head costs several times what noise costs; removing the bulk costs the same as noise. A window's bulk share, from one forward pass, detects recitation without labels.

A theory (Section 4). Two Taylor orders of the token loss with the population Fisher as the only object predict these laws and the outcome of edits. The collateral of block deletions and of matched noise is predicted within 2–10% in OLMo-2 7B and OLMo-3 7B with identical code and no free parameter; inside the bulk the cost depends only on the energy moved. Forgetting is first order at a logit margin, which predicts which population a deletion reaches, and the coupling of a token to a layer factors into two energies, which yields a layer map whose parameter-free prediction of margin-shift variance holds within a factor two in both models. A minimum-interference argument prices the storage of an item by the inverse product of its whitened energies, which says why flat items are cheap to store and predicts the two-sidedness of the whitening; a causal experiment on OLMo-2 1B (Appendix E) shows the optimizer setting where a fresh imprint goes, the representation recoding toward the bulk under every optimizer, and continued population training erasing head-written content first.

Consequences for editing (Section 5). Because every label-free edit inside the bulk pays the same price per unit of energy moved, better curvature cannot beat the identity there: the frontier is set by energy and reachable with the eigenbasis alone. Two levers move it. Placement by the layer map cut the recall cost per unit of forgetting by $2.3\times$ where $1.4\times$ was predicted. A coherent step along a memorized set's shared direction, built from half the set, removes the other half at a tenth of the ordinary-text cost of deletion, and reaches the late boilerplate that no bulk deletion reaches (86–91% of the held-out half at norm 2.5 in both 7B models, at half a percent of a nat), where gradient-ascent and preference-based unlearning at matched forgetting cost –. The collateral of a coherent edit is the alignment of its direction with the preserved population.

2 Setting

Curvature of a layer. Fix a linear layer $y = Wa$, $W \in \mathbb{R}^{O\times I}$, inside a network with token loss ℓ_t. Write $g_t = \partial\ell_t/\partial y_t$ for the output gradient and a_t for the input at token t. For a population $\mathcal{D}$ of tokens the Kronecker factors are $A = \mathbb{E}_{\mathcal{D}}[aa^\top]$ and $B = \mathbb{E}_{\mathcal{D}}[gg^\top]$, with eigenbases P (eigenvalues μ_i) and Q (eigenvalues λ_o); coefficients are $C = Q^\top WP$, $\tilde{a} = P^\top a$, $\tilde{g} = Q^\top g$. We use the *coupled* factors, in which each side is weighted by the other's squared norm, and sampled labels, so that the per-coefficient curvature $\Lambda_{oi} = \mathbb{E}_{\mathcal{D}}[\tilde{g}_o^2\tilde{a}_i^2]$ is the diagonal of the Fisher in this basis [George et al., 2018]. The *depth* of an output direction is $d_o = \sum_i \Lambda_{oi}$, and symmetrically for inputs. *Head* and *bulk* are the top and bottom of the depth ordering; the bulk block of a matrix is the flattest $60\%\times 60\%$ of directions and the head block the top $20\% \times 20\%$, unless stated otherwise. The joint first-order logit shift of a perturbation Δ of several matrices at token t is $s_t(\Delta) = \sum_m g_{t,m}^\top \Delta W_m a_{t,m}$.

Models, populations, edits. The primary model is OLMo-2 7B [Team, 2024], edited at the gate and up projections of layers 23–25 of 32; the reference population is 1152 sequences of 512 tokens from its mid-training mix [AI2, 2024]. The memorized set is 1054 windows of 112 tokens from a deduplicated Dolma split that the model recites (prefix 64, greedy suffix 48), so its provenance is known; ordinary text is the reference's never-recited windows; out-of-distribution text is the Pile; facts are the answer tokens of 316 NaturalQuestions and PopQA items the unedited model answers correctly. For any model a *recitation scan* of the 9216 reference windows yields a memorized population without ground truth: the windows recited verbatim (129 for OLMo-2 7B, 127 for OLMo-2 1B, 111 for OLMo-3 7B, 21 at ≥ 0.95 suffix accuracy for Qwen2.5-7B, which was not trained on this corpus). Edits are block deletion ($\Delta = -C_S$ on a block S, possibly scaled by α), noise of matched Frobenius norm in the same block, and coherent steps along a chosen direction. Forgetting is the fraction of items still recited (all 48 suffix tokens argmax-correct); collateral is the mean loss increase in nats on ordinary and out-of-distribution text; capabilities are a 12-task suite (closed-book recall, reading comprehension, commonsense, arithmetic) evaluated with olmo-eval.

3 Where memorized content sits

Head and bulk. In layers 4–28 of OLMo-2 7B the coupled curvature spectrum of each gate and up projection has a near-degenerate bulk and a small head: the middle 80% of directions span a factor 3–5 in depth and the directions above ten times the median depth are 0.1–0.5% of the total; the same holds in OLMo-3 7B (Table 1). Within the bulk the depth *ordering* is reproducible (rank correlation 0.99 between halves of the reference data) but the eigenvectors are not: two references agree on which directions are high-gain (rank correlation 0.89–0.98) while their bulk eigen-subspaces overlap barely above chance. A bulk direction is not canonical; its gain is. The bulk is not rarely used either (effective number of contributing sequences ~ 1500 of 2304 against 1750–1940 for the head): the head is gain, not breadth. Bulk plus outliers is the familiar shape of Hessian spectra [Sagun et al., 2017, Papyan, 2019]; here it is resolved per layer and per side.

Empirical law 1 (Removal versus noise)**.** Deleting the bulk block of a layer costs ordinary text what noise of the same norm in the same block costs; deleting the head block costs several times more than noise.

On the quintile blocks of OLMo-2 7B the deletion-to-noise cost ratio is $0.73, 0.99, 0.95, 0.91$ on the four bulk quintiles and 4.4 on the head quintile; on the bulk and head blocks of every model and population in Table 1 it is 1.0–1.5 and 2.5–11. Inside the bulk the shape of a perturbation is irrelevant: seven noise shapes (isotropic, inverse-curvature, inverse-magnitude, small- or large-components only, rank 32, an oracle ratio) cost the same at norms 90 and 120, while across the four bulk quintiles the same norm costs $+0.0013$ to $+0.0041$ nats, graded by the block's mean curvature. The attention projections of the same layers behave alike: their head blocks cost 60–100$\times$ more to remove than their bulk blocks, and the recited windows' input-side profile at the query projection is whitened at -0.3 in both 7B models (Appendix G).

Empirical law 2 (Half-whitening on both sides)**.** Per direction of the reference eigenbasis, a pretraining-era memorized population's mean activation energy and mean output-gradient energy fall relative to ordinary text's as a power of the reference energy, with the same exponent on the two sides.

For the Dolma set of OLMo-2 7B the exponent of $\log(E_{\mathrm{mem}}/E_{\mathrm{ord}})$ on $\log E_{\mathrm{ord}}$ is -0.65 to -0.69 on the input side ($R^2 = 0.94$) and -0.62 to -0.69 on the output side of the up projection ($R^2 = 0.85$); clean and typical text give -0.09 to $+0.22$ and a split-half null $+0.01$ (Figure 1). The population's total activation energy is normal (0.87–0.95$\times$) but spread far more evenly: $2.75\times$ the ordinary energy in the flattest fifth of input directions, $0.6\times$ in the head, steepest reallocation in the middle. The share of *curvature* along an output direction, which marginalises one side against the other, inherits an exponent near $-1/2$. Fact answer tokens have the same input-side profile as memorized suffix tokens (bulk shares 0.45 and 0.48; exponents -0.73 to -0.81): flatness marks item-specific content in general.

Empirical law 3 (Flatness is a property of the population, and it predicts deletability)**.** Within one model, populations memorized at different times sit at different depths of the spectrum, and a population's bulk-share shift predicts the fraction of its recitation that deleting a bulk block removes.

Table 1 and Figure 2 place the memorized populations of four models on common axes. Two populations live in OLMo-2 7B: the 1054 Dolma windows, half of them recited verbatim by the 1.26T-token

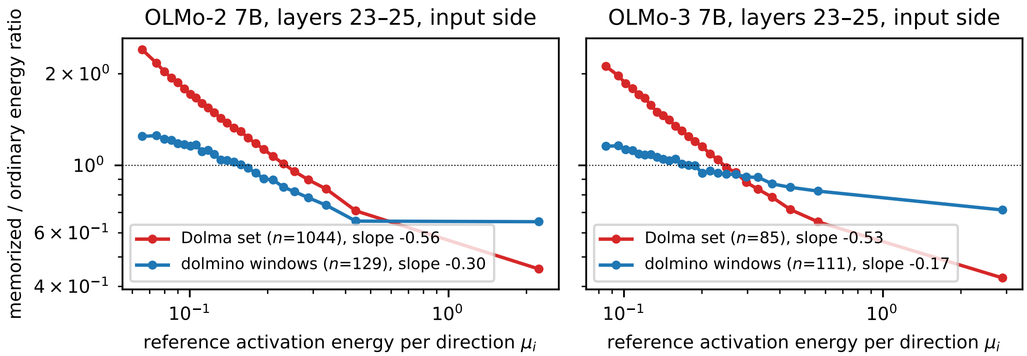

Figure 1: Whitening profiles. Ratio of the memorized population's mean activation energy per direction to ordinary text's, against the reference energy of the direction (binned over the input eigenbasis of the first matrix of the band, 25 bins), for the Dolma set and the recited dolmino windows in OLMo-2 7B and OLMo-3 7B. Slopes are least-squares exponents over all directions. The pretraining-era Dolma populations are whitened at -0.55 to -0.67; the late boilerplate at -0.2 to -0.33.

Table 1: Six memorized populations in four models, late MLP band (layers 23–25 of 32 for the 7B models, 11–13 of 16 for the 1B, 20–22 of 28 for Qwen). Whitening exponent on the input side; bulk share of activation energy for ordinary and recited text; the two-halves cosine of the population's bulk direction; the product-law coupling ratio to ordinary text; recitation remaining after deleting the bulk block, with the ordinary-text cost; and the removal-to-noise cost ratios on the head and bulk blocks. Populations come from the recitation scan except the Dolma sets.

model	population (n)	exponent	bulk share	halves cos	coupling	bulk deletion	head/bulk ratio
OLMo-2 7B	Dolma set (1054)	-0.67	$0.19 \to 0.45$	0.64	$\sim$24	0.18, +0.024	4.4 / 1.0
OLMo-2 7B	dolmino windows (129)	-0.33	$0.27 \to 0.37$	0.88	15	0.89, +0.020	11 / 1.3
OLMo-3 7B	Dolma set (85)	-0.55	$0.27 \to 0.47$	0.10	33	0.40, +0.010	6.1 / 1.2
OLMo-3 7B	dolmino windows (111)	-0.20	$0.27 \to 0.34$	0.89	8	0.955, +0.010	6.7 / 1.3
OLMo-2 1B	dolmino windows (127)	-0.34	$0.19 \to 0.31$	0.90	29	0.00, +0.070	4.6 / 1.5
Qwen2.5-7B	dolmino windows (21)	-0.52	$0.19 \to 0.43$	(0.09)	58	0.33, +0.025	2.5 / 1.3

checkpoint and two thirds by the end of pretraining, and the 129 dolmino windows it recites, which its checkpoints show were memorized in the 50B-token mid-training that follows (7, 8, 8, 10 recited at 1.26, 2.1, 2.9 and 3.8T tokens). The dolmino windows are globally duplicated boilerplate: 107 are also recited by OLMo-3 7B, 118 by OLMo-2 1B, 10 by Qwen2.5-7B. The Dolma set is whitened at -0.67, moves 26 points of its input energy into the bulk, and deleting the bulk block of layers 23–25 leaves 18% of it recited at $+0.024$ nats; the dolmino windows are whitened at -0.33, move 10 points, and the same deletion leaves 89% of them at $+0.020$, while only removing the head block, at ten times the cost, forgets them. OLMo-3 7B reproduces the split on the same two populations: its 85 recited Dolma windows (-0.55, $R^2 = 0.88$, $+20$ points) go from 0.99 to 0.40 recited under the bulk deletion at $+0.010$ nats; its 111 recited dolmino windows (-0.20, $+7$ points) go from 1.00 to 0.955 under the identical edit. Across models, populations and bands the bulk-share shift orders the forgotten fraction (Figure 2, left), and the first-order margin theory of Section 4 predicts it from a quarter-deletion probe (right). Flatness is not a property of models, nor of verbatim text as such, but of how and when a population was stored; Section 4.3 says why storage depth should track the training that follows memorization.

A forward pass detects memorized windows. The bulk share of a window's suffix activations at the input of one matrix, from one forward pass and one fixed basis, separates recited from ordinary windows with no labels (**??**): AUC – over the 9216 reference windows of the two 7B models, with partly recited windows in between. It is a detector of memorization that needs neither generation nor provenance.

Empirical law 4 (One direction per memorized distribution)**.** The summed bulk-projected margin gradients of two disjoint halves of a memorized distribution have cosine 0.6–0.9; those of two

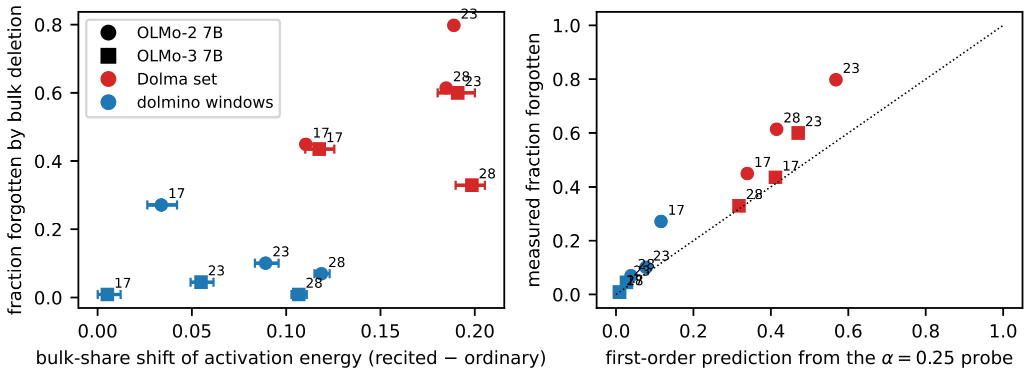

Figure 2: Deletability. Left: fraction of a population's recitation removed by deleting the bulk block of a three-layer band, against the shift of the population's activation bulk share relative to ordinary text (bootstrap 95% intervals over items), for two models, two populations and three bands (labels give the first layer of the band). Right: the same fraction against the first-order prediction from a quarter-deletion probe (Proposition 2).

memorized distributions have cosine ≈ 0; and each is nearly orthogonal to the summed margin gradients of ordinary text and of facts.

For the Dolma set the halves' cosine is 0.62–0.65 in the bulk (0.74–0.92 unprojected); for the 129 self-labelled recited windows it is 0.86 in the bulk (0.91 unprojected), and their cosine with the Dolma direction is 0.01. The cosine of the Dolma direction with ordinary text's is 0.004 and with facts' 0.001 in the bulk. Two-halves cosines of 0.90 (OLMo-2 1B) and 0.89 (OLMo-3 7B) replicate the law on the recited windows; for Qwen2.5-7B the population is too small to test it (0.09 at ten items per half), and OLMo-3's 85 recited Dolma windows, a scattering of individually memorized items rather than a set memorized together, share little (0.10). The shared component belongs to populations memorized as a set, with a common source, format and phase. Within a set the per-item directions are mostly orthogonal (median pairwise cosine 0.002) with a coherent minority that clusters into groups (Section 5).

Empirical law 5 (Learning flattens what it memorizes, by recoding)**.** An item's profile flattens on both sides only when training fits it to saturation, and it does so by redistributing a conserved representational energy from the head to the bulk.

Between the 1.26T-token checkpoint of OLMo-2 7B and the final model, the 112 ordinary windows that become recited move from a typical profile ($b = +0.08$ input, $+0.24$ output) to a flat one (-0.17, -0.52); the Dolma set, already recited at 1.26T, is flat at both checkpoints. Across all 9216 windows the bulk share of input energy is 0.18–0.19 for any final recitation accuracy below 0.9, 0.257 in $[0.9, 1)$ and 0.273 at 1.0: a threshold at saturated fit, the same threshold the margin picture puts on memorization. Over the same interval ordinary text *sharpens*, its bulk fraction falling $0.32 \to 0.23$ and head fraction rising $0.40 \to 0.50$ as shared computation consolidates, at $1.30\times$ the total energy. The windows that become recited are exempt and slightly reverse ($0.29 \to 0.30$ bulk, $0.42 \to 0.38$ head) at conserved energy ($1.07\times$). Memorization exempts an item from the consolidation into the head that ordinary text undergoes, and moves it a little further out; no energy is added.

Empirical law 6 (One direction per memorized distribution)**.** The summed bulk-projected margin gradients of two disjoint halves of a memorized set have cosine 0.6–0.9; those of two memorized sets have cosine ≈ 0; each is nearly orthogonal to the summed margin gradients of ordinary text and of facts.

Halves' cosines are 0.62–0.65 for the Dolma set and 0.86–0.90 for the recited windows of OLMo-2 7B, 1B and OLMo-3 7B; the two sets' directions have cosine 0.01, and the Dolma direction's cosine with ordinary text's is 0.004 and with facts' 0.001. OLMo-3's 85 recited Dolma windows, a scattering of individually memorized items rather than a set memorized together, share little (0.10): the shared component belongs to populations memorized as a set. Within a set the per-item directions are mostly orthogonal (median pairwise cosine 0.002) with a coherent minority that clusters into groups (Section 5).

Empirical law 7 (Learning flattens what it memorizes, by recoding)**.** An item's profile flattens only when training fits it to saturation, and it does so by redistributing a conserved representational energy from the head to the bulk.

Between the 1.26T-token checkpoint and the final OLMo-2 7B, the 112 ordinary windows that become recited move from a typical profile ($+0.08$ input, $+0.24$ output) to a flat one (-0.17, -0.52). Across all 9216 windows the bulk share of input energy is 0.18–0.19 for any final recitation accuracy below 0.9, 0.257 in $[0.9, 1)$ and 0.273 at 1.0: a threshold at saturated fit. Over the same interval ordinary text *sharpens* (bulk fraction $0.32 \to 0.23$, head $0.40 \to 0.50$, at $1.30\times$ the energy) while the windows that become recited are exempt and slightly reverse ($0.29 \to 0.30$, $0.42 \to 0.38$) at conserved energy ($1.07\times$). Memorization exempts an item from the consolidation into the head that ordinary text undergoes; no energy is added. Appendix E reproduces the recoding on a fresh imprint under three optimizers.

4 A theory in two orders

The expansions below are standard: a second-order Taylor expansion with the Gauss–Newton Hessian underlies optimal brain damage [LeCun et al., 1989, Hassibi et al., 1993] and influence functions [Koh and Liang, 2017, Grosse et al., 2023], and the population Fisher as the metric for moving weights is natural gradient [Amari, 1998, Martens, 2020] and elastic weight consolidation [Kirkpatrick et al., 2017]. What is new is the claim that they hold *quantitatively and without calibration* for the edits and populations of Section 3, which of their components fail and how, and what they imply for where memorized content sits and what an edit can do. Proofs are one line each (Appendix B).

4.1 Collateral

Proposition 1 (Two-order collateral)**.** *For a population $\mathcal{D}$ and a perturbation Δ, $\Delta L_{\mathcal{D}} = \mathbb{E}_{\mathcal{D}}[s_t] + \frac{1}{2}\,\mathbb{E}_{\mathcal{D}}^{\text{sampled}}[s_t^2] + R_3$, where the first expectation uses true-label gradients, the second uses labels sampled from the model (the Fisher form), and R_3 is third order.*

Measured per token, the form predicts the collateral of bulk removal and of matched bulk noise on ordinary and Pile text within 2–10% in OLMo-2 7B and OLMo-3 7B with identical code and no calibration (Figure 5, Table 7), and within 12% for the deletions of two references' bulk blocks in OLMo-2 (Table 4). Four corollaries carry the empirical content.

Corollary 1 (Shape invariance and graded price)**.** *For a perturbation with independent zero-mean coefficients of variance σ_{oi}^2, $\mathbb{E}[\Delta L_{\mathcal{D}}] = \frac{1}{2}\sum_{oi}\Lambda_{oi}\sigma_{oi}^2$ exactly. On a block where Λ is constant this is $\frac{1}{2}\bar{\Lambda}\|\sigma\|^2$: the cost depends on the energy moved and on nothing else, and across blocks it is graded by $\bar{\Lambda}$.*

This is Empirical law 1's shape invariance and quintile grading; the diagonal form with EK-FAC's Λ reproduces all ten band measurements with log-space $R^2 = 0.97$ and one fitted constant of 0.57 against the theoretical $1/2$.

Corollary 2 (Removal versus noise)**.** *Removing a block S and adding noise of the same norm cost, to second order and neglecting off-diagonal fourth moments, in the ratio $\langle\Lambda\rangle_{C^2}/\langle\Lambda\rangle_{\text{unif}}$: one where Λ is constant (the bulk), more where the stored coefficients sit on the high-curvature components (the head).*

Predicted 1.0 on the bulk quintiles (measured 0.73–0.99) and 2.8 on the head (measured 4.4). Head removal is super-quadratic in both 7B models: the quadratic form under-predicts its cost by $1.5\times$ in OLMo-3 and $3.7\times$ in OLMo-2, at logit shifts of 3.6, while it predicts head *noise* to within 10% (Figure 5). The excess is the operational content of "content versus computation."

Corollary 3 (Out of distribution, interference, relativity, factorisation)**.** *(i) Since $|\ell(z+\delta) - \ell(z)| \le 2\|\delta\|_\infty$ for cross-entropy, the quadratic form is an upper bound at large logit shifts: the deletions of Table 4 cost the Pile 1.75–$2\times$ less than predicted at rms shifts of 0.4 logits. (ii) The first-order term* *of removal is the directional derivative of the population's loss along $-C_S$, and it is negative in every measurement (30–40% of the quadratic term in OLMo-3): stored content interferes, and deleting it relieves. (iii) "Private" is a two-place relation: deleting the Pile-defined bulk costs dolmino $3\times$ what its own deletion costs. (iv) The diagonal approximation $\mathbb{E}[\tilde{g}_o\tilde{g}_{o'}\tilde{a}_i\tilde{a}_{i'}] \approx \delta_{oo'}\delta_{ii'}\Lambda_{oi}$ is exact for random perturbations and under-predicts coherent removals by 2–2.5×, because C_S is the population's own accumulated gradient structure.*

4.2 Forgetting at the margin

Let $m_t = z_{t,y_t} - \max_{j\neq y_t} z_{t,j}$ be the logit margin of a recited token; greedy recitation holds while $\min_t m_t > 0$. Restricting $\nabla_W m_t$ to a block S, call $c_t = \nabla_S m_t \cdot C_S$ the block's *coherent contribution* and $e_t = \|\nabla_S m_t\|^2$ the token's *energy* in the block.

Proposition 2 (Removal and noise at the margin)**.** *Removal of S shifts m_t by $-c_t$ to first order. Isotropic noise of per-coefficient variance σ^2 shifts it by a random variable of standard deviation $\sigma\sqrt{e_t}$ and mean $\frac{1}{2}\sigma^2\,\mathrm{tr}_S\,\nabla_W^2 m_t \leq 0$.*

On the Dolma set of OLMo-2 7B (Table 5) the coherent contribution equals the noise spread rather than exceeding it, so the "coherent N against incoherent $\sqrt{N}$" intuition is false here; the noise mean is quadratic in the spread; the shrink response is linear in α to $\alpha \approx 0.5$; and one perturbation moves an item's 48 tokens independently (within-item correlation 0.01). On the dolmino windows of both 7B models the response is linear to $\alpha = 1$ and the first-order prediction of strict recitation from a quarter-deletion probe is within one to three points at every α (Table 7): these windows survive because their margins are 12–13 logits and the block's coherent contribution is 1–1.5, where the Dolma set's margins move by 2.9 against minimum margins near 2.3. The systematic mean shift under noise is not the gate's convexity nor either normaliser's (each was built into a model and refuted, Appendix C); it is the second-order response of the later layers, with one constant measured per model.

Corollary 4 (Selectivity and the coupling law)**.** *For two populations of equal alignment, the one coupling k times more strongly suffers k times more from removal and between k and k^2 times more from noise, so noise is at least as selective as removal. To the product approximation the coupling factors, $e_t \approx (\sum_{o\in S}\tilde{g}_{to}^{m\,2})(\sum_{i\in S}\tilde{a}_{ti}^2)$, so the ratio of two populations' noise-shift variances is the ratio of their products of bulk energies.*

Memorized against fact tokens in OLMo-2 7B: coherent contributions 2.92 against 1.38, spreads 3.17 against 1.65, means 2.19 against 0.82, with equal alignments to 10%; on the capability suite, at matched forgetting, bulk noise costs 5 points of closed-book recall where deletion costs 8. The product law predicts 3.04 against 3.7 measured for memorized against facts, 15.2 against 18.1 for recited against ordinary windows in OLMo-2, and 7.5 against 14.5 in OLMo-3; in every case the ratio is carried by the output side (input-side ratios 1.0–1.1): the populations are identical on the way in and differ in how much of their margin passes through the layer. Summing the coupling over positions and layers gives a *layer map* (Figure 4); isotropic noise of per-entry variance σ^2 on a band's matrices then has predicted margin-shift variance $\sigma^2\sum_l C_l$, which holds within a factor two for recited and ordinary tokens in nine of ten bands of the two 7B models, and within 10–30% on the ratio in eight.

4.3 The storage cost of an item

Proposition 3 (Minimum-interference storage)**.** *Let an item require a first-order margin gain $\sum_{oi} M_{oi}\Delta_{oi} \geq \delta$, with M its summed margin gradient in the reference eigenbasis. The perturbation of least population collateral $\frac{1}{2}\sum\Lambda_{oi}\Delta_{oi}^2$ is $\Delta_{oi}^* \propto M_{oi}/\Lambda_{oi}$, the natural gradient of the margin, and its cost is $\frac{1}{2}\delta^2/\sum_{oi} M_{oi}^2/\Lambda_{oi} \approx \frac{1}{2}\delta^2/[(\sum_o \tilde{g}_o^2/\lambda_o)(\sum_i \tilde{a}_i^2/\mu_i)]$: the inverse product of the item's* whitened *energies on the two sides.*

Hence (i) items whose energies sit in flat directions are cheap to store, and a learner that trades fit against collateral stores them preferentially and stores them there; (ii) the two sides enter symmetrically, which is the two-sidedness of Empirical law 2; (iii) at fixed energy the cost falls when energy moves toward the bulk, so learning that lowers the cost *redistributes*, adding nothing: Empirical law 7; (iv) the readout is the coupling law's product, so the same energies measure fragility (an

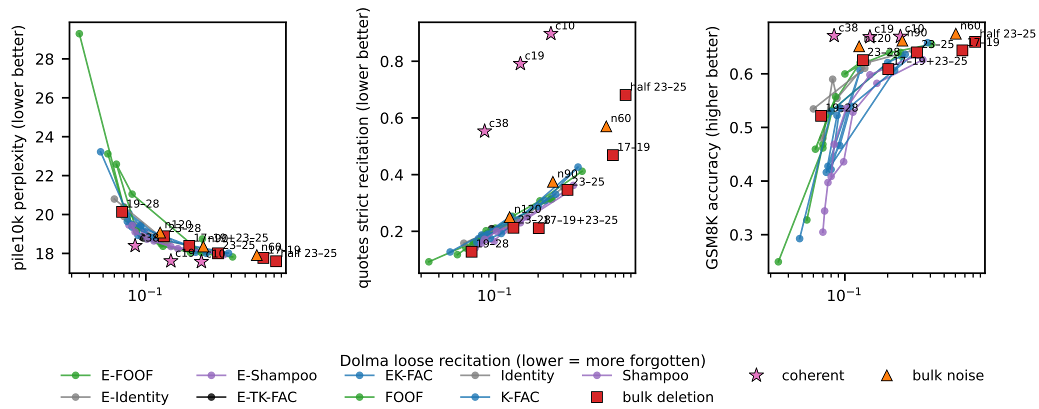

Figure 3: The curvature-editing frontier of Anonymous [2026] and our edits on its axes (OLMo-2 7B, layers 23–25 unless labelled, Dolma memorized set). Lines: K-FAC, FOOF, Shampoo and identity metrics, with (E-) and without eigenvalue corrections, over the retention threshold ρ, calibrated on the dolmino mix. Markers: bulk deletions (labelled by band), bulk noise (norm), the coherent step along the Dolma set's direction (norm), and gradient-ascent and NPO unlearning at matched forgetting. Perplexity on pile10k, strict recitation of web-duplicated quotes, and GSM8K accuracy, against loose recitation of the Dolma set.

item's bulk share predicts the shrink factor at which it stops being recited, Spearman -0.44) and detection (**??**).

Corollary 5 (Selective decay)**.** *Under gradient flow on the population loss, linearised around its optimum, a perturbation component along an eigen-direction of curvature κ decays as $e^{-\eta\kappa t}$ (as $e^{-\eta P(\kappa)\kappa t}$ under a preconditioner P). Continued training on the population is a low-pass filter on whatever has been written into the weights.*

This is the weight-side half of the mechanism behind Empirical law 3: a population memorized early is filtered, one memorized at the end of training is not. Appendix E tests it on fresh imprints in OLMo-2 1B: SGD writes a head-heavy imprint (input-side exponent $+0.7$) that fifty steps of population training erase, Adam's has five times more bulk energy and survives, the natural gradient's is in the bulk by construction (exponents -0.5 to -0.9, the profile of the pretraining-era populations) and costs the population a third of Adam's collateral; under every optimizer the items' representation recodes toward the bulk ($-0.08 \to -0.24$ to -0.33) at conserved energy. But the natural-gradient imprint decays as fast as Adam's, and in every decay run the recoded representation drifts back as recitation is lost: the population's training undoes the *key* as well as the imprint. Persistence needs recurrence, and what survives a long stretch of training is the part of both that the population's gradient leaves alone. The exponent of the whitening is a measured constant per population, not a derived one (Appendix D).

5 Consequences for editing

Why curvature fidelity cannot matter inside the bulk. The methods compared in Anonymous [2026] (K-FAC, EK-FAC, E-FOOF, E-Shampoo, identity, with and without eigenvalue corrections and saliency ranking) differ in how well they estimate Λ. By Corollary 1 the cost of any label-free edit inside the bulk is $\frac{1}{2}\bar{\Lambda}$ times the energy it moves, whatever its shape, and by Section 4.2 the forgetting it buys is set by the same energy. Inside the bulk the estimate cannot matter, and it does not: the methods' frontiers coincide, and plain deletion of the bulk block, needing one collection pass and no corrections, ranking or calibration, lands on the best of them (six-layer deletion against EK-FAC at matched Dolma recitation: $+0.48$ perplexity for $+0.047$ GSM8K). Every label-free edit we built moves along that frontier, not beyond it (Figure 3): deletions and noise at other norms and bands trade perplexity against quotes and GSM8K on the same surface the curvature methods trace. The coherent edit alone leaves it. What the curvature estimate does buy is the head/bulk boundary and the depth ordering, which every approximation finds.

Table 2: Placement by the layer map in OLMo-2 7B. Deletion of the bulk block at three bands; capability changes in points on the suite; recall per unit forgotten is the summed NaturalQs and PopQA loss divided by the forgotten fraction of the memorized set.

deletion	forgotten	NaturalQs	PopQA	HellaSwag	recall per unit forgotten
layers 17–19	0.45	−1.8	−2.2	−2.0	4.4
layers 18–20	0.53	−3.7	−3.0	−2.3	6.3
layers 23–25	0.79	−7.9	−8.0	−1.3	10.1

Table 3: Unlearning the recited dolmino windows at matched held-out forgetting, editing the gate and up projections of layers 23–25 only: the coherent step, gradient ascent with a retain loss, and NPO. Recitation of the held-out half, loss change on ordinary and Pile text, and capability-suite changes (points).

–

Lever one: placement. The layer map predicts that an edit is most selective where the target population's coupling exceeds the preserved population's by the most; in OLMo-2 7B the memorized-to-fact ratio peaks in layers 17–20. It predicted that a bulk deletion at 17–19 would cost $1.4\times$ less closed-book recall per unit of forgetting than at 23–25, and spare arithmetic; the measurement is $2.3\times$ (Table 2), at the price of two points of HellaSwag.

Lever two: coherence. Empirical law 6 implies an edit: a coherent step along the bulk-projected margin gradient of a memorized set, built from one half and scored on the other. On the Dolma set of OLMo-2 7B at norm 19 the held-out half falls from 0.98 to 0.16 recited, matching bulk deletion's 0.18, at $+0.002$ nats on ordinary text against $+0.024$, $+0.000$ on the Pile against $+0.027$, and $-0.7/0.0$ points of NaturalQs/PopQA against $-7.9/-8.0$ (Table 6); its perplexity through the benchmark pipeline is 17.61 against 17.62 unedited, a point no label-free edit approaches. On the late boilerplate that no bulk deletion reaches, a step of norm 2.5 removes 86–91% of the held-out half in both 7B models at $+0.005$ to $+0.007$ nats with the Pile untouched (Table 3), where the same forgetting by gradient ascent with a retain loss or by negative preference optimization [Zhang et al., 2024a] on the same matrices costs –. Two of four predictions written before the experiment failed informatively: *weak transfer* (items never used fall to 16%, because the set shares a direction) and *shape invariance* (the direction's collateral per unit norm is 2–5$\times$ that of energy-matched noise: it sits at the high-curvature end of the bulk).

The collateral is alignment; grouping buys choice. The Dolma direction removes 84% of held-out Dolma recitation at norm 19 and 5% of the recited windows'; the recited windows' own direction removes 86% of their held-out half at norm 5 and 2.5% of the Dolma set's, but costs ordinary dolmino text $+0.045$ nats, some $600\times$ the per-energy cost of bulk noise, because its cosine with the direction that raises ordinary dolmino margins is 0.30 (the Dolma direction's is 0.004). For every coherent target population there is a mean bulk direction that labels, or the model's own recitation, reveal and no label-free edit can find; its reach is the population that shares it, and its collateral is the alignment of that direction with the preserved population. Within the recited windows, average linkage on per-item directions finds two super-groups; a direction from one removes all of its recitation at norm 5 and leaves the other at 0.88–1.00, at the same collateral per item as the union (0.00037 against 0.00039 nats): clean selectivity, no discount. A label-free procedure follows: detect (**??**), group, and remove coherently what is foreign to what is to be preserved.

6 The theory's tests on a second model

Table 1 carries the laws across four models: the head/bulk asymmetry, the bulk-ward shift of recited text's activation energy, the shared direction per memorized set and the coupling law replicate wherever the population is large enough, and the whitening exponent is a property of the population. The theory's quantitative tests were then run with the same code, and no tuning, on OLMo-3 7B with OLMo-2 7B as the control (Table 7, Figures 4 and 5). Collateral: bulk removal and noise within 2–10% on ordinary and Pile text in both models, head removal super-quadratic in both. Forgetting: first-order recitation from the quarter-deletion probe within one to three points; the two-coefficient

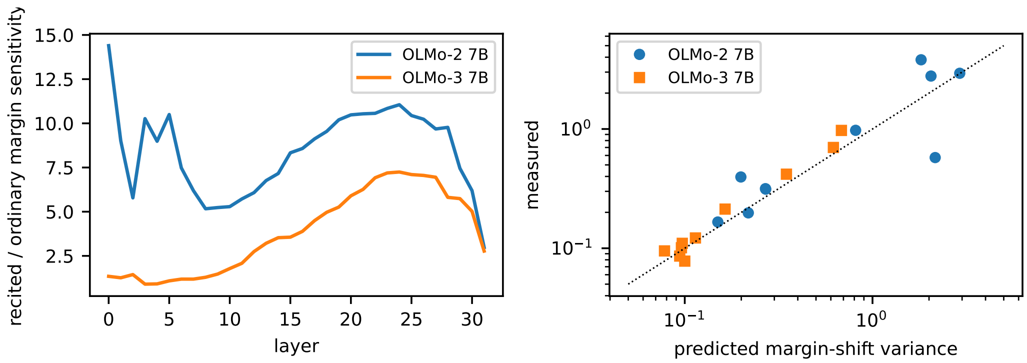

Figure 4: The layer map and its test. Left: ratio of recited to ordinary margin sensitivity per layer (Section 4.2) in both 7B models. Right: predicted against measured margin-shift variance under isotropic noise of per-matrix norm 60 on five three-layer bands, recited and ordinary tokens, no free parameter.

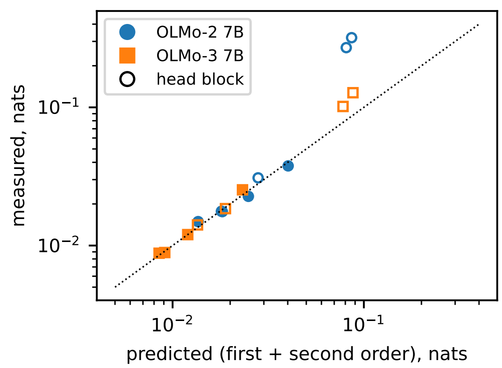

Figure 5: Two-order collateral (Proposition 1): predicted against measured loss change on ordinary and Pile text for bulk removal, bulk noise, head removal and head noise in both 7B models. Bulk edits and head noise lie on the diagonal; head removal (open markers) is super-quadratic.

Gaussian model for noise, fitted at three norms, over-predicts forgetting at the fourth by 5 and 12 points, the saturation of the mean shift seen before. Coupling: 15.2 against 18.1 and 7.5 against 14.5. Placement: the map's absolute variance within a factor two in nine of ten bands, failing only at layers 3–5 of OLMo-2, where twenty-seven layers of nonlinearity separate the perturbation from the logits. At matched norm OLMo-3's margins move three times less than OLMo-2's for both populations with the same ratio: matched norm is not matched effect across models, and the map is what makes them comparable.

7 Related work

Memorization and its location. Verbatim memorization scales with model size, duplication and prompt length [Carlini et al., 2023, Nasr et al., 2023], and is tied to the long tail [Feldman, 2020, Feldman and Zhang, 2020]. Attempts to localize memorized paragraphs in weights or neurons [Stoehr et al., 2024, Chang et al., 2024, Huang et al., 2024] report distributed, partly shared machinery; Hase et al. [2023] find that causal localization does not predict where editing works. Our location is in a different coordinate system, the curvature spectrum of a layer, where memorized content is not a set of parameters but a spectral profile, and where location across layers (coupling) rather than across the spectrum is what predicts editing outcomes. Merullo et al. [2025] propose the low-curvature intuition that this paper turns into laws and a theory, and Anonymous [2026] benchmark the curvature approximations whose coincidence we explain.

Unlearning and forgetting. Gradient ascent on the forget set with a retain term [Jang et al., 2023], negative preference optimization [Zhang et al., 2024a], representation misdirection [Li et al., 2024] and approximate unlearning by fine-tuning [Eldan and Russinovich, 2023, Maini et al., 2024] are the standard baselines; Section 5 compares the coherent edit with the first two at matched forgetting. Selective forgetting through the Fisher was proposed by Golatkar et al. [2020]. The decay of what a population's gradient overwrites is the classical account of catastrophic interference [McCloskey and Cohen, 1989, Kirkpatrick et al., 2017, Ramasesh et al., 2021]; Corollary 5 is its statement in the curvature eigenbasis, and Appendix E its test on a fresh imprint. Memorization dynamics during pretraining, including that larger models memorize faster and that memorization can precede overfitting, are studied by Tirumala et al. [2022] and Biderman et al. [2023]; our checkpoints add that the spectral location of a memorized population depends on when it was memorized.

Curvature and its approximations. K-FAC [Martens and Grosse, 2015], its eigenvalue correction [George et al., 2018], Shampoo [Gupta et al., 2018] and their unifying view [Eschenhagen et al., 2023] supply the eigenbases we use; the Fisher as a metric for the cost of moving weights goes back to natural gradient [Amari, 1998, Martens, 2020] and to its use for protecting old knowledge [Kirkpatrick et al., 2017], of which Proposition 3 is the storage-side reading. Influence functions in the EK-FAC metric [Grosse et al., 2023, Koh and Liang, 2017, Bae et al., 2024] and pruning by second-order saliency [LeCun et al., 1989, Hassibi et al., 1993] share the two-order expansion; the failure of factorised curvature for coherent perturbations (Section 4.1) is the same off-diagonal structure that makes influence estimates sensitive to their Hessian [Hong et al., 2025]. The bulk-plus-outliers shape of the Hessian spectrum is known for small networks [Sagun et al., 2017, Papyan, 2019, Ghorbani et al., 2019] and, blockwise, for transformers [Zhang et al., 2024b], as is the concentration of gradient descent in the outlier subspace [Gur-Ari et al., 2018]; that item-specific content lives in the bulk while shared computation lives in the outliers is, to our knowledge, new.

8 Limitations and conclusion

The laws are established in the late MLP band of one model with a memorized population of known provenance, and replicated in three others with populations the models reveal by recitation; attention weights, embeddings and the first and last layers, whose spectra differ, are not treated. The exponent of the whitening is a measured constant per model, not a derived one, and the systematic mean shift under noise carries one measured constant. The two-order theory owns the regime of small logit shifts: deletion beyond half a block and noise beyond the norms probed leave it, and out of distribution it gives a bound rather than a value. The coherent edit needs a labelled or self-labelled set and removes what that set shares, no more and no less.

Within those limits the picture is simple. A trained matrix has a head that computes and a bulk that stores. Memorized text ends up in the bulk because the bulk is where storing it costs the population least and because whatever was written elsewhere the population overwrites; how far a population has gone down that road is what decides whether a bulk deletion can reach it, and its own shared direction reaches it either way. The same population metric prices every edit: energy moved times curvature for label-free edits, coupling times margin for forgetting, alignment with the preserved population for coherent edits. Better curvature cannot lower the label-free price. Choosing the layer and the direction can.

References

AI2. Dolmino Mix 1124. `https://huggingface.co/datasets/allenai/dolmino-mix-1124`, 2024.

Shunichi Amari. Natural gradient works efficiently in learning. *Neural Computation*, 10(2):251–276, 1998.

Anonymous. Hessian approximations beyond optimization, 2026. Companion manuscript, under review.

Juhan Bae, Wu Lin, Jonathan Lorraine, and Roger Grosse. Training data attribution via approximate unrolling. In *Advances in Neural Information Processing Systems*, 2024.

Stella Biderman, USVSN Sai Prashanth, Lintang Sutawika, Hailey Schoelkopf, Quentin Anthony, Shivanshu Purohit, and Edward Raff. Emergent and predictable memorization in large language models. In *Advances in Neural Information Processing Systems*, 2023.

Nicholas Carlini, Daphne Ippolito, Matthew Jagielski, Katherine Lee, Florian Tramèr, and Chiyuan Zhang. Quantifying memorization across neural language models. In *International Conference on Learning Representations*, 2023.

Ting-Yun Chang, Jesse Thomason, and Robin Jia. Do localization methods actually localize memorized data in llms? a tale of two benchmarks. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics*, 2024.

Ronen Eldan and Mark Russinovich. Who's harry potter? approximate unlearning in LLMs. *arXiv preprint arXiv:2310.02238*, 2023.

Runa Eschenhagen, Alexander Immer, Richard E Turner, Frank Schneider, and Philipp Hennig. Kronecker-factored approximate curvature for modern neural network architectures. In *Thirty-seventh Conference on Neural Information Processing Systems*, 2023.

Vitaly Feldman. Does learning require memorization? a short tale about a long tail. In *Proceedings of the 52nd Annual ACM SIGACT Symposium on Theory of Computing*, 2020.

Vitaly Feldman and Chiyuan Zhang. What neural networks memorize and why: Discovering the long tail via influence estimation. In *Advances in Neural Information Processing Systems*, 2020.

Thomas George, César Laurent, Xavier Bouthillier, Nicolas Ballas, and Pascal Vincent. Fast approximate natural gradient descent in a kronecker-factored eigenbasis. In *Neural Information Processing Systems*, 2018.

Behrooz Ghorbani, Shankar Krishnan, and Ying Xiao. An investigation into neural net optimization via hessian eigenvalue density. In *International Conference on Machine Learning*, 2019.

Aditya Golatkar, Alessandro Achille, and Stefano Soatto. Eternal sunshine of the spotless net: Selective forgetting in deep networks. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020.

Roger Grosse, Juhan Bae, Cem Anil, Nelson Elhage, Alex Tamkin, Amirhossein Tajdini, Benoit Steiner, Dustin Li, Esin Durmus, Ethan Perez, et al. Studying large language model generalization with influence functions. *arXiv preprint arXiv:2308.03296*, 2023.

Vineet Gupta, Tomer Koren, and Yoram Singer. Shampoo: Preconditioned stochastic tensor optimization. In *International Conference on Machine Learning*, 2018.

Guy Gur-Ari, Daniel A. Roberts, and Ethan Dyer. Gradient descent happens in a tiny subspace. *arXiv preprint arXiv:1812.04754*, 2018.

Peter Hase, Mohit Bansal, Been Kim, and Asma Ghandeharioun. Does localization inform editing? surprising differences in causality-based localization vs. knowledge editing in language models. In *Advances in Neural Information Processing Systems*, 2023.

Babak Hassibi, David G Stork, and Gregory J Wolff. Optimal brain surgeon and general network pruning. In *IEEE international conference on neural networks*, pages 293–299. IEEE, 1993.

Dat Minh Hong, Bruno Kacper Mlodozeniec, Runa Eschenhagen, and Richard E. Turner. Better hessians matter: Studying the impact of curvature approximations in influence functions. In *Mechanistic Interpretability Workshop at NeurIPS 2025*, 2025.

Jing Huang, Diyi Yang, and Christopher Potts. Demystifying verbatim memorization in large language models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, 2024.

Joel Jang, Dongkeun Yoon, Sohee Yang, Sungmin Cha, Moontae Lee, Lajanugen Logan IV, and Minjoon Seo. Knowledge unlearning for mitigating privacy risks in language models. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*, 2023.

Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations*, 2015.

James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, Joel Veness, Guillaume Desjardins, Andrei A. Rusu, Kieran Milan, John Quan, Tiago Ramalho, Agnieszka Grabska-Barwinska, Demis Hassabis, Claudia Clopath, Dharshan Kumaran, and Raia Hadsell. Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, 114(13):3521–3526, 2017.

Pang Wei Koh and Percy Liang. Understanding black-box predictions via influence functions. In *International Conference on Machine Learning*, 2017.

Yann LeCun, John S Denker, and Sara A Solla. Optimal brain damage. In *Advances in Neural Information Processing Systems*, volume 2, 1989.

Nathaniel Li, Alexander Pan, Anjali Gopal, Summer Yue, Daniel Berrios, Alice Gatti, Justin D. Li, Ann-Kathrin Dombrowski, Shashwat Goel, Long Phan, et al. The WMDP benchmark: Measuring and reducing malicious use with unlearning. In *International Conference on Machine Learning*, 2024.

Pratyush Maini, Zhili Feng, Avi Schwarzschild, Zachary C. Lipton, and J. Zico Kolter. TOFU: A task of fictitious unlearning for LLMs. In *Conference on Language Modeling*, 2024.

James Martens. New insights and perspectives on the natural gradient method. In *Journal of Machine Learning Research*, volume 21, pages 1–76, 2020.

James Martens and Roger Grosse. Optimizing neural networks with kronecker-factored approximate curvature. In *International Conference on Machine Learning (ICML)*, 2015.

Michael McCloskey and Neal J. Cohen. Catastrophic interference in connectionist networks: The sequential learning problem. *Psychology of Learning and Motivation*, 24:109–165, 1989.

Jack Merullo, Srihita Vatsavaya, Lucius Bushnaq, and Owen Lewis. From memorization to reasoning in the spectrum of loss curvature. *arXiv preprint arXiv:2510.24256*, 2025.

Milad Nasr, Nicholas Carlini, Jonathan Hayase, Matthew Jagielski, A. Feder Cooper, Daphne Ippolito, Christopher A. Choquette-Choo, Eric Wallace, Florian Tramèr, and Katherine Lee. Scalable extraction of training data from (production) language models. In *arXiv preprint arXiv:2311.17035*, 2023.

Vardan Papyan. Measurements of three-level hierarchical structure in the outliers in the spectrum of deepnet hessians. In *International Conference on Machine Learning*, 2019.

Vinay V. Ramasesh, Ethan Dyer, and Maithra Raghu. Anatomy of catastrophic forgetting: Hidden representations and task semantics. *International Conference on Learning Representations*, 2021.

Levent Sagun, Utku Evci, V. Ugur Güney, Yann Dauphin, and Léon Bottou. Empirical analysis of the hessian of over-parametrized neural networks. *arXiv preprint arXiv:1706.04454*, 2017.

Niklas Stoehr, Mitchell Gordon, Ryan Cotterell, and Owen Lewis. Localizing paragraph memorization in language models. *arXiv preprint arXiv:2403.19851*, 2024.

OLMo Team. 2 OLMo 2 Furious. *arXiv preprint arXiv:2501.00656*, 2024.

Kushal Tirumala, Aram H. Markosyan, Luke Zettlemoyer, and Armen Aghajanyan. Memorization without overfitting: Analyzing the training dynamics of large language models. *Advances in Neural Information Processing Systems*, 2022.

Ruiqi Zhang, Licong Lin, Yu Bai, and Song Mei. Negative preference optimization: From catastrophic collapse to effective unlearning. In *Conference on Language Modeling*, 2024a.

Yushun Zhang, Congliang Chen, Tian Ding, Ziniu Li, Ruoyu Sun, and Zhi-Quan Luo. Why transformers need Adam: A Hessian perspective. In *Advances in Neural Information Processing Systems*, 2024b.

Table 4: Two-order prediction of the collateral of block deletions in OLMo-2 7B (layers 23–25, nats per token). In distribution the sum is within 12% of the measurement; out of distribution the quadratic form is an upper bound (Corollary 3) and the first-order term is negative (Section 4.1).

edit $\to$ population	first order	second order	sum	measured
dolmino bulk $\to$ dolmino	-0.0025	$+0.0209$	$+0.0184$	$+0.021$
Pile bulk $\to$ dolmino	-0.0040	$+0.0607$	$+0.0567$	$+0.063$
dolmino bulk $\to$ Pile	-0.0228	$+0.0784$	$+0.0556$	$+0.027$
Pile bulk $\to$ Pile	-0.0147	$+0.0410$	$+0.0263$	$+0.015$

Table 5: Per-token margin shifts of the memorized suffix tokens of OLMo-2 7B under removal (shrink) of the bulk block and noise of matched norm.

norm	removal mean (s.d.)	noise mean (s.d.)	$\lvert\text{mean}_{\text{rem}}\rvert/\text{s.d.}_{\text{noise}}$	noise mean/variance
19	-0.20 (0.72)	-0.14 (0.66)	0.30	-0.32
38	-0.76 (1.60)	-0.56 (1.41)	0.54	-0.28
57	-1.72 (2.64)	-1.27 (2.29)	0.75	-0.24
76	-2.92 (3.70)	-2.19 (3.17)	0.92	-0.22

A Tables

B Proofs

Proposition 1. Write $\ell_t(\Delta) = \ell(z_t + \delta z_t(\Delta))$ with z_t the logits and δz_t their change. To second order in Δ, $\delta z_t = J_t\Delta + O(\Delta^2)$ and $\ell_t(\Delta) - \ell_t(0) = \nabla_z \ell^\top J_t\Delta + \frac{1}{2}(J_t\Delta)^\top \nabla_z^2 \ell\,(J_t\Delta) + \nabla_z\ell^\top(\text{second-order part of } \delta z_t) + O(\Delta^3)$. The Gauss–Newton approximation drops the last term. For softmax cross-entropy $\nabla_z\ell = p - e_y$ and $\nabla_z^2\ell = \operatorname{diag}(p) - pp^\top = \mathbb{E}_{y'\sim p}[(e_{y'} - p)(e_{y'} - p)^\top]$, so $(J_t\Delta)^\top\nabla_z^2\ell(J_t\Delta) = \mathbb{E}_{y'\sim p}[((e_{y'} - p)^\top J_t\Delta)^2]$, the second moment under sampled labels of the first-order loss shift $s_t = (e_{y'} - p)^\top J_t\Delta$. Taking the population mean gives the statement. For a perturbation of a linear layer $y = Wa$, $(e_{y'} - p)^\top J_t\Delta = g_t^\top \Delta W a_t$ summed over the perturbed matrices, which is $s_t(\Delta)$ as defined in Section 2.

Corollary 1. $\mathbb{E}[s_t^2] = \sum_{oi,o'i'} \Delta_{oi}\Delta_{o'i'}\tilde{g}_{to}\tilde{g}_{to'}\tilde{a}_{ti}\tilde{a}_{ti'}$; for independent zero-mean coefficients the expectation over Δ kills all terms with $(oi) \neq (o'i')$ and leaves $\sum_{oi}\sigma_{oi}^2\mathbb{E}[\tilde{g}_o^2\tilde{a}_i^2] = \sum_{oi}\sigma_{oi}^2\Lambda_{oi}$. The first-order term has zero mean.

Corollary 2. With the diagonal approximation, removal costs $\frac{1}{2}\sum_S \Lambda_{oi}C_{oi}^2$ and noise of the same norm $\|C_S\|^2$ spread uniformly over $|S|$ coefficients costs $\frac{1}{2}(\|C_S\|^2/|S|)\sum_S \Lambda_{oi}$; the ratio is as stated.

Corollary 3. $\ell(z) = -z_y + \operatorname{lse}(z)$; $\nabla\operatorname{lse}$ is a probability vector, so lse is 1-Lipschitz in $\|\cdot\|_\infty$ and $|\ell(z+\delta) - \ell(z)| \leq |\delta_y| + \|\delta\|_\infty \leq 2\|\delta\|_\infty$.

Proposition 2. First order in Δ: $\Delta m_t = \nabla_S m_t \cdot \Delta$, which is $-c_t$ for $\Delta = -C_S$ and a zero-mean variable of variance $\sigma^2\|\nabla_S m_t\|^2 = \sigma^2 e_t$ for isotropic noise. Second order: $\mathbb{E}[\frac{1}{2}\Delta^\top\nabla_W^2 m_t\Delta] = \frac{1}{2}\sigma^2\operatorname{tr}_S\nabla_W^2 m_t$. The margin is $z_y - \max_{j\neq y} z_j$; the maximum is convex in z, so $-\max$ contributes a non-positive trace through the logit Hessian, and the target logit of a fitted token is locally concave in the weights producing it in the measured regime.

Proposition 3. Minimise $\frac{1}{2}\sum_{oi}\Lambda_{oi}\Delta_{oi}^2$ subject to $\sum_{oi} M_{oi}\Delta_{oi} = \delta$. The Lagrangian condition is $\Lambda_{oi}\Delta_{oi} = \theta M_{oi}$, so $\Delta_{oi}^* = \theta M_{oi}/\Lambda_{oi}$ with $\theta = \delta/\sum_{oi} M_{oi}^2/\Lambda_{oi}$, and the cost is $\frac{1}{2}\theta^2\sum_{oi} M_{oi}^2/\Lambda_{oi} = \frac{1}{2}\delta^2/\sum_{oi} M_{oi}^2/\Lambda_{oi}$. For a single token $M_{oi} = \tilde{g}_o\tilde{a}_i$ and, with $\Lambda_{oi} \approx \lambda_o\mu_i$, the sum factors as $(\sum_o \tilde{g}_o^2/\lambda_o)(\sum_i \tilde{a}_i^2/\mu_i)$. In the full parameter space the same argument gives $\Delta^* = F^{-1}\nabla m$, the natural gradient of the margin.

Table 6: The coherent edit in OLMo-2 7B: a step along the bulk-projected summed margin gradient of half the memorized set. Recitation of the train and held-out halves, collateral in nats, arithmetic accuracy.

edit	train half recited	held-out half recited	ordinary	Pile	arithmetic
coherent, norm 19	0.019	0.156	$+0.002$	$+0.000$	0.892
coherent, norm 38	0.013	0.055	$+0.014$	$+0.022$	0.887
coherent, norm 76	0.008	0.013	$+0.106$	$+0.143$	0.838
bulk deletion, norm 76	0.180		$+0.024$	$+0.027$	0.857
bulk noise, norm 90	0.159		$+0.026$	$+0.047$	0.868

Table 7: Parameter-free predictions on OLMo-3 7B and OLMo-2 7B with identical code (layers 23–25; bulk block of norm 84.5 and 76.2; recited dolmino windows). Top: two-order collateral (Proposition 1) of bulk removal and matched noise, predicted from the first-order logit shift by central differences in float32, against measured, in nats per token. Middle: forgetting (Proposition 2). Bottom: the layer map's predicted margin-shift variance under isotropic noise of per-matrix norm 60 on five three-layer bands, for recited and ordinary tokens, against measured.

Collateral: edit → population; first order, second order, sum; measured			
OLMo-2 7B	bulk remove → ordinary	-0.0052	$+0.0233$
OLMo-2 7B	bulk remove → Pile	-0.0058	$+0.0460$
OLMo-2 7B	bulk noise → ordinary	-0.0012	$+0.0148$
OLMo-2 7B	bulk noise → Pile	-0.0002	$+0.0251$
OLMo-3 7B	bulk remove → ordinary	-0.0058	$+0.0143$
OLMo-3 7B	bulk remove → Pile	-0.0018	$+0.0250$
OLMo-3 7B	bulk noise → ordinary	$+0.0003$	$+0.0088$
OLMo-3 7B	bulk noise → Pile	-0.0000	$+0.0121$
Forgetting: strict recitation predicted against measured			
OLMo-2 7B	shrink $\alpha = 0.5$ / 0.75 / 1	predicted 0.969 / 0.961 / 0.922	measured 0.961 / 0.9
OLMo-3 7B	shrink $\alpha = 0.5$ / 0.75 / 1	predicted 0.991 / 0.982 / 0.982	measured 0.991 / 0.9
OLMo-2 7B	noise, two-coefficient model, norm 19 / 38 / 57 / 76	predicted 0.994 / 0.967 / 0.897 / 0.767	measured 0.988 / 0.973 /
OLMo-3 7B	noise, two-coefficient model, norm 21 / 42 / 63 / 85	predicted 1.000 / 0.996 / 0.975 / 0.925	measured 1.000 / 1.000 /
OLMo-2 7B	coupling law, variance ratio recited/ordinary	predicted 15.2	measured 18.
OLMo-3 7B	coupling law, variance ratio recited/ordinary	predicted 7.5	measured 14.
Placement: band; predicted variance (recited, ordinary); measured; ratio predicted / measured			
OLMo-2 7B	3–5	2.16, 0.22	0.58, 0.20
OLMo-2 7B	9–11	0.81, 0.15	0.98, 0.17
OLMo-2 7B	16–18	1.82, 0.20	3.81, 0.40
OLMo-2 7B	22–24	2.92, 0.27	2.93, 0.31
OLMo-2 7B	28–30	2.05, 0.27	2.78, 0.32
OLMo-3 7B	3–5	0.10, 0.10	0.11, 0.08
OLMo-3 7B	9–11	0.16, 0.09	0.21, 0.09
OLMo-3 7B	16–18	0.35, 0.08	0.42, 0.10
OLMo-3 7B	22–24	0.68, 0.10	0.97, 0.10
OLMo-3 7B	28–30	0.62, 0.11	0.70, 0.12

C The systematic shift under noise

Proposition 2 leaves the negative mean shift as a concavity. Three mechanisms were tested by constructing the model each implies and reading off the margins, so that a refutation is a refutation and not a poor fit. (i) *Convexity of the gate*: noise on the gate pre-activation has mean effect equal to replacing SiLU by its Gaussian smoothing; the smoothed-gate model shifts margins by $+0.002$ to $+0.005$ where -0.14 and -0.56 are measured at norms 19 and 38. (ii) *The final normaliser*: under norm-76 noise the final residual rms changes by 0.09% and the logit scale not at all. (iii) *The edited layers' own normalisers*: OLMo-2 adds $\mathrm{RMSNorm}(\mathrm{mlp}(h))$ to the residual, and noise inflates the MLP output variance by 40–48% at layers 23–25 (predicted to first order and confirmed), shrinking each layer's genuine contribution by 16–18%; but the noise-free model with those outputs scaled by the measured factors moves the mean margin by -0.02 against -2.19 measured. Scaling a layer's output is a different operation from removing its block. What remains is the zero-mean

Table 8: Fresh imprints in OLMo-2 1B (layers 9–13, 96 items, aggressive regime: eight item windows and four reference sequences per step at equal weight). Exponent of the imprint's energy per direction against the population eigenvalue (positive: head-heavy); the imprint's energy share in the bulk block (0.36 if uniform); ratio of head- to bulk-block imprint energy; recitation after 50 steps of reference-only training with the same optimizer; the items' input-side whitening exponent at layer 13 before and after memorization.

optimizer	imprint exponent G / A	bulk share	head/bulk	steps to recite	recited after 50 ref. steps
SGD	$+0.35$ to $+0.62$ / $+0.65$ to $+0.71$	0.07–0.13	3.6	225	$0.97 \to 0.08$
Adam	$+0.11$ to $+0.44$ / $+0.51$ to $+0.67$	0.14–0.22	0.75	325	$0.96 \to 0.42$, back to 0.94 a
natural gradient	-0.65 to -1.08 / -0.47 to -0.88	0.54–0.69	0.05	500	$0.99 \to 0.22$ under SGD, $0.99 \to 0.4$

part of the perturbation entering the residual stream and the second-order response of layers 26–31 to it, $\mathbb{E}[\Delta m_t] = \frac{1}{2}\operatorname{tr}(H_t\Sigma_t)$, which yields the exactly quadratic scaling (constant to 3% over four norms), the proportionality to the token's coupling (correlation 0.56 with the removal shift) and equal population coefficients relative to margin; its constant, $-\mathbb{E}[\Delta m]/\operatorname{Var}[\Delta m] \approx 0.22$–$0.32$ per logit, is measured once per model.

D On the exponent

The profile $E_{\mathrm{mem},i}/E_{\mathrm{ord},i}$ against $E_{\mathrm{ord},i}$ is monotone with local exponents between -0.4 and -0.9 and flattest at both ends of the spectrum. An additive item drive against a curvature-proportional restoring force under plain gradient descent gives $E_{\mathrm{mem},i} = c\mu_i + F/(r\mu_i)$, a U-shaped ratio flat in the head and $\propto \mu^{-2}$ in the bulk, which the monotone measurement excludes. Optimizer preconditioning points the right way: Adam [Kingma and Ba, 2015] normalises each coordinate's update by the square root of its second moment, which in the population eigenbasis is $\propto \sqrt{\lambda_o\mu_i}$, so item-specific content accumulates with an extra $\kappa^{-1/2}$ relative to gradient descent, which is the exponent of the curvature share; and Proposition 3 says that the exact natural-gradient step whitens fully. The measured representational exponent lies between the two, varies across models, and the drift of the weights across checkpoints (the flattest decile drifts only 10–20% more than the sharpest) shows that bulk directions are not slowly learned. What sets the exponent is therefore the joint recoding of the item's representation by the earlier layers and the imprint in the later ones, under the population's restoring force; writing that down is the open derivation.

E A causal test: writing a fresh imprint

Proposition 3 makes claims about learning, and learning can be run. We memorized 96 never-recited dolmino windows into OLMo-2 1B by training the gate and up projections of layers 9–13 on mixed batches of item windows and reference sequences until the items were recited, with three optimizers on the same data: SGD with momentum, whose step is the item gradient; Adam; and the natural gradient in the population K-FAC basis, $\Delta W = -\eta\, Q[(Q^\top G P) \oslash (\lambda\mu^\top + \delta)]P^\top$, which is the minimum-interference direction of Proposition 3 by construction. The imprint $W - W_0$ is then read in the population eigenbasis, the items' activation profile at the inputs of layers 11–13 is compared before and after, and the band is trained further on reference sequences alone to see what survives.

Three things happen (Table 8). *The optimizer sets where the imprint goes.* SGD's imprint follows the item gradient: its energy per direction grows with the population eigenvalue (exponent $+0.65$ to $+0.71$ on the input side, $R^2 = 0.97$), only 7–13% of it lies in the bulk block, and removing the imprint's bulk block leaves 92% of the items recited while removing its head block removes all of them. Adam's imprint has five times more bulk energy at the same recitation, and the natural gradient's is in the bulk by construction: its energy per direction falls with the eigenvalue at exponents -0.5 to -1.1 (R^2 0.93–0.99), 54–69% of it lies in the bulk block, removing that block removes the items (0.06 recited) and removing the head block leaves them (0.92). The natural-gradient imprint's profile is the profile of the pretraining-era populations of Table 1; the SGD and Adam imprints have the head-heavy profile of no memorized population we have measured. At matched recitation the natural gradient cost the reference population $+0.07$ nats against Adam's $+0.15$–0.21 and SGD's $+0.19$, and cost the Pile as much as they did ($+0.20$): the minimum-interference direction is minimum for the population that defined it (Section 4.1(ii), now causal). *The head-stored imprint does not survive the population, but bulk storage alone does not persist either.* Fifty steps of training on reference sequences alone erase the SGD imprint's recitation ($0.97 \to 0.08$, item loss $0.04 \to 0.8$) while the bulk-heavier Adam imprint dips and returns (0.42 at 50 steps, 0.94 at 200, 0.82 at 400); with the phase-two optimizer exchanged the ordering SGD-imprint $<$ Adam-imprint holds (under SGD the Adam imprint keeps 0.48 recited at item loss 0.30 where the SGD imprint keeps 0.09 at 0.7; under Adam the Adam imprint keeps 0.9 at 0.06 where the SGD imprint keeps 0.6 at 0.19). Content written into the head is content the population's gradient acts on, and it is overwritten first. But the natural-gradient imprint, in the bulk by construction, decays as fast as Adam's or faster ($0.99 \to 0.22$ under SGD, $0.99 \to 0.44$ under Adam), and in every decay run the items' bulk share at layer 13 falls back toward the ordinary value ($0.32 \to 0.27$ against $0.24 \to 0.21$) as recitation is lost. What the population's training undoes is not only the imprint but the *key*: the recoded representation that reads it. An item persists while imprint and key are re-fit together, that is, while it recurs; what survives a long stretch of training after the item stops recurring is the part of both that the population's gradient leaves alone. Corollary 5 is that filter on the weight side; the representation side, which this experiment exposes, is the missing half, and it is why a population memorized in the last 50B tokens (Empirical law 3) has a different profile from one memorized in the first trillion and re-fit for the rest of training. *The representation recodes, whatever the optimizer.* Before memorization the items' activation profile at the inputs of layers 11–13 is typical (exponent -0.08); after it, it is whitened, increasingly with the number of trained layers upstream ($-0.18, -0.27, -0.33$ at layers 11, 12, 13 under SGD; $-0.12, -0.21, -0.29$ under Adam; $-0.09, -0.19, -0.29$ under the natural gradient, or -0.24 at layer 13 against ordinary text measured on the same final model), at conserved total energy (0.86–$1.06\times$). The items' bulk share tracks the fit and falls back as they are forgotten. Learning flattens what it memorizes even when the weight imprint that does the memorizing is head-heavy: the flattening is in the representation the earlier trained layers hand on, as Empirical law 7 found in the pretraining run. The regime is aggressive, with items at 8% of the training tokens, and the 1152-sequence reference pool overfits after two epochs, so the collateral figures of this section compare optimizers at matched steps rather than measure the cost of memorization; a gentler regime on a 35k-sequence pool with an item-free control is reported in Appendix F.

F The gentle regime

Items at one quarter of the loss weight, four item windows and eight reference sequences per step from a pool of $34{,}795$ sequences (under one epoch), with an item-free control run of the same optimizer and length. Adam reaches 0.97 recited in 650 steps with the same imprint profile as in the aggressive regime (input-side exponents $+0.50$ to $+0.58$, bulk share 0.17–0.22) and the same recoding of the items' representation at layer 13 (-0.24 against ordinary text on the same final model, bulk share 0.315 against 0.225, energy 0.98–$1.05\times$); removing the imprint's head block leaves 0.03 recited and removing its bulk block 0.84. SGD reaches 0.98 in 1100 steps with input-side imprint exponents $+0.65$ to $+0.77$ and bulk share 0.07–0.13, recodes the items to -0.27 at layer 13 (0.326 against 0.219), and again stores the recitation in the imprint's head block (0.00 left when it is removed, 0.84 when the bulk block is). The optimizer sets the imprint's location and the regime does not. The item-free Adam control run improves held-out ordinary text by 0.0155 nats over the same 650 steps (in-distribution fine-tuning) and leaves the Pile unchanged (-0.0016), while the run that also memorizes the 96 items ends at $+0.0123$ on ordinary text and $+0.0181$ on the Pile: net of the control, memorizing 96 windows of 112 tokens with Adam costs the reference population 0.028 nats per token and the Pile 0.020, about what deleting the bulk block of three layers costs in the 7B models. The natural gradient reaches 0.97 in 2450 steps with the bulk imprint of the aggressive run (input-side exponents -0.60 to -0.89, bulk share 0.59–0.70; removing the imprint's bulk block leaves 0.06 recited, its head block 0.90) and the same recoding (-0.26 at layer 13). Its collateral peaks at $+0.012$ nats on ordinary text midway and ends at -0.005 on ordinary text and $+0.000$ on the Pile, against the control's -0.016 and -0.002 at 650 steps: net of the control, at most $+0.011$ and $+0.002$, a third and a tenth of Adam's, with the caveat that the natural gradient took four times as many steps, and the population training between exposures is part of what repairs the damage. Written where the population's curvature is lowest, the items cost the population least, as Proposition 3 says, and the repair happens while the items keep recurring.

G Attention projections

The probe of Section 3 run on the query, key, value and output projections of layers 23–25 (coupled bases from the same reference data). Recited dolmino windows, input side of the query projection: exponent -0.32 in OLMo-2 7B and -0.29 in OLMo-3 7B (bootstrap 95% intervals $[-0.34, -0.30]$ and $[-0.30, -0.27]$), bulk share 0.35 against 0.19 for ordinary text; output projection -0.03 and -0.24. Removing the bulk block of the four projections (norm 38–41 per matrix) leaves 0.96 and 1.00 of the windows recited at $+0.003$ and -0.000 nats on ordinary text; removing the head block (norm 19–24) leaves 0.23 and 0.27 at $+0.25$ and $+0.17$. The removal-to-noise asymmetry and the location of the late boilerplate outside the bulk hold for attention as for the MLPs.

H Experimental details

Factors and bases. Coupled K-FAC factors are accumulated over 1152 reference sequences of 512 tokens with labels sampled from the model, $A_k \mathrel{+}= \sum_t \|g_t\|^2 a_t a_t^\top$ and $B_k \mathrel{+}= \sum_t \|a_t\|^2 g_t g_t^\top$ per matrix, and diagonalised in float32. Directions are ordered by eigenvalue; the depth ordering by $\sum_i \Lambda_{oi}$ agrees with it to rank correlation above 0.98. Block edits act on the outer product of the chosen output and input directions.

Recitation scan. Each of the 9216 windows of 112 tokens is prompted with its first 64 tokens and decoded greedily for 48; a window is recited if all 48 tokens match, ordinary if fewer than 75% match. Recitation of an edited model is scored by teacher-forced argmax agreement on the 48 suffix tokens, which coincides with greedy decoding.

Two-order predictions. The first-order logit shift of an edit is computed by central differences in float32 at ± 0.02 of the edit; the first-order loss term is $(p - e_y)^\top \delta z$ and the second-order term $\frac{1}{2}(\sum_y p_y \delta z_y^2 - (p^\top \delta z)^2)$, averaged over tokens; the measured change applies the full edit.

Layer map. For each layer, $C_l = \frac{1}{N} \sum_s \|a_{l,s}\|^2 \sum_t (\|\partial m_t / \partial h_{l,s}\|^2 + \|\partial m_t / \partial u_{l,s}\|^2)$ over target tokens t and all positions s, with h, u the gate and up pre-activations, estimated by backpropagating $\sum_t \varepsilon_t m_t$ for random signs ε_t (4 draws). Isotropic noise of per-entry variance σ^2 on a band's gate and up matrices then has predicted margin-shift variance $\sigma^2 \sum_{l \in \text{band}} C_l$.

Capability suite. Twelve tasks from olmo-eval with capped instance counts: NaturalQuestions and PopQA (closed-book recall), DROP, SQuAD and Jeopardy (reading comprehension and recall), HellaSwag, ARC, OpenBookQA, WinoGrande, CommonsenseQA (commonsense), GSM8K and a curated arithmetic set.